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FAIRGAME

A framework for AI agent bias recognition using game theory

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Why predicting AI agents is hard

More agents, less predictability

Letting AI agents interact adds complexity to interpreting and predicting their outcomes.

Theory without good tools

Game theory models strategic interaction, but it needs reproducible, standardized frameworks to test it.

A shared framework

FAIRGAME simulates games with LLM agents and compares results with game-theoretic predictions.

How FAIRGAME works

STEP 1

Configuration file

JSON file with agents, personalities, payoff matrix, rounds and languages.

STEP 2

Prompt template

Game narrative in any language, filled in each round with payoffs and history.

STEP 3

LLM agents

Agents built from any LLM through its API, so models can be compared directly.

STEP 4

Standard outputs

Full game histories in a consistent format, ready for analysis.

Agents can differ in personality, strategic attitude, language and cultural orientation. FAIRGAME is open source, so simulations can be reproduced and compared.

What the experiments found

4
LLMs tested

5
languages

2
classic games

72,000
individual decisions

Prisoner's Dilemma	Mostly defection, as game theory predicts, but results shift with language and personality.
Battle of the Sexes	Mostly cooperative, but selfish personas make cooperation drop sharply. Claude 3.5 Sonnet reaches near-perfect coordination.
Identity test	Disguised games were still recognized as known dilemmas, so LLMs rely on prior knowledge, not only payoffs.

Scoring LLMs and next steps

Metric	What it measures
Internal variability	Consistency when the same game is replayed
Cross-language inconsistency	Stability of results across the five languages
Sensitivity to payoff	Response to harsh versus mild payoff changes
Variability over rounds	How much strategies fluctuate from round to round

Standout results

Llama 3.1 405B is the most stable overall in the Prisoner's Dilemma.

Mistral Large shows the greatest inconsistency across languages.

Next steps

Test nationality, gender and age biases, add agent communication, and scale beyond two agents.